

THE EVOLUTION OF AI: Emerging Agentic Strategies and the Road Ahead

From AI Agents to AGI and ASI — A Plain-Language Guide
to the Most Important Technology Shift of Our Time

- **RAJ SANGAM**

7

Evolution
Stages

6+

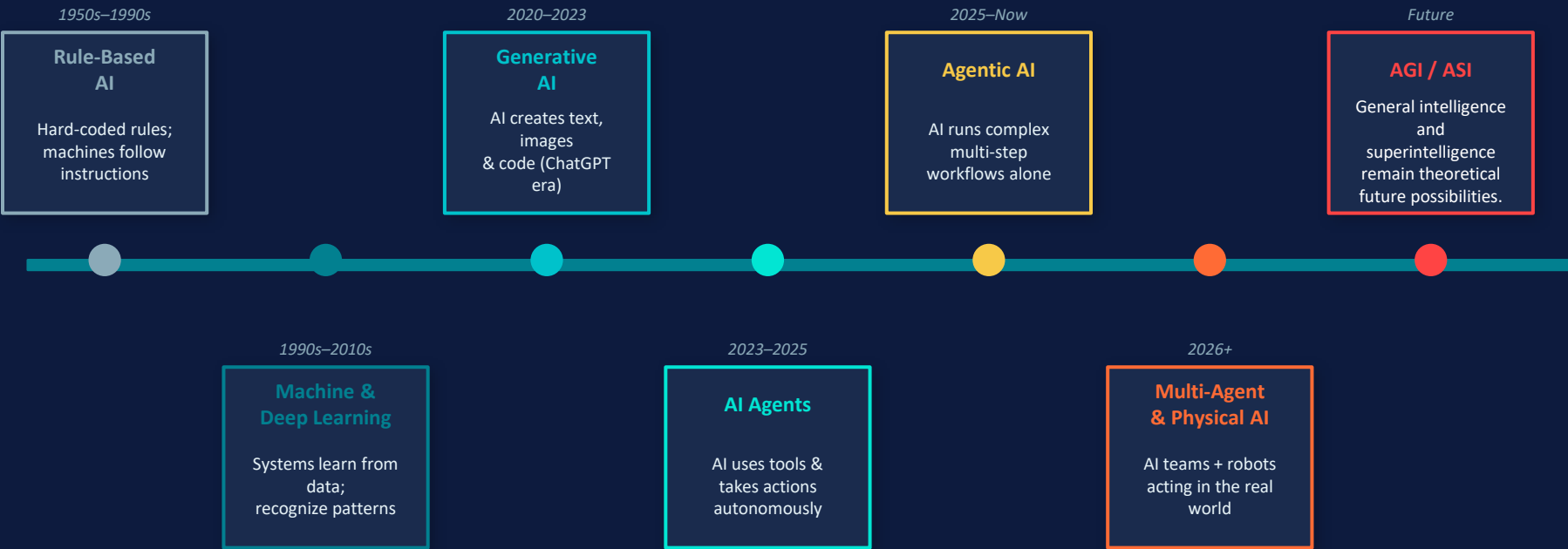
Agentic
Strategies

2026

Current
Horizon

AI Evolution Timeline

FROM SIMPLE RULES TO AUTONOMOUS INTELLIGENCE — HOW WE GOT HERE



◀ You Are Here — Stage 3 → 4

THINK OF AN AI AGENT AS A SMART ASSISTANT THAT CAN TAKE ACTIONS, NOT JUST ANSWER QUESTIONS



What is an AI Agent?

An AI Agent is like a smart digital worker. You give it a goal — like "book me a flight" or "summarize these reports" — and it figures out the steps to get there on its own.

It can search the web, read files, call other systems, write code, and check its own work — all without you needing to guide it step by step.

The big difference from a regular chatbot: an agent takes actions in the world, not just provides answers.



Memory

Remembers what happened earlier



Tool Use

Calls APIs, searches web, runs code



Retrieval

Looks up facts from docs & databases



Reasoning

Thinks step-by-step to solve problems

Real-World Agent Examples



Claude Code (Anthropic)

You describe a coding task in plain English — e.g., "add login with Google to my app." Claude Code reads your entire codebase, writes the code, runs tests to check it works, fixes any errors it finds, and prepares the change for review. No manual step-by-step needed.



ChatGPT Operator / Agent

An AI agent that can actually use a web browser like a person would. You say "order me pizza from my usual place" and it opens the browser, navigates the website, fills in your address, and places the order — seeing and clicking just like you would.



Microsoft Copilot (M365)

Built into Microsoft Office apps, Copilot can read your emails, join your meetings, and write reports based on your data. Ask it "summarize last week's emails about the budget" and it searches your inbox, pulls the relevant threads, and gives you a clean summary.

AI THAT HANDLES ENTIRE WORKFLOWS FROM START TO FINISH — NOT JUST SINGLE TASKS



What is Agentic AI?

Imagine assigning a project to a capable employee — not just asking one question. Agentic AI takes a high-level goal like "prepare a competitive analysis report" and handles everything: planning the steps, gathering information, writing the draft, checking it for errors, and delivering the final output.

It's the difference between a calculator (answers one thing) and an employee (manages a whole project).

1 · Plan

2 · Execute

3 · Reflect

4 · Adapt

💡 How Agents become Agentic AI: A single AI Agent handles one task. Agentic AI chains multiple agents and steps together into a full workflow — like an assembly line where each station hands off to the next.

Agentic AI in Action



Customer Support Automation

A customer asks about an insurance claim. The system checks their identity, pulls their claim history, calculates what they owe, writes a personalized reply, and escalates to a human only if needed — all in seconds. What used to need 3 departments now runs automatically.



Travel Planner Agent

You say "Plan a 5-day trip to Japan in October, budget \$3,000." The agent searches flights and hotels, checks your calendar, compares prices, builds an itinerary, books everything, and sends you a confirmation — in minutes, not hours.



Research & Report Assistant

Give it a research question. It searches dozens of sources, reads the relevant ones, pulls out key findings, checks for contradictions, and writes a structured summary report — replacing hours of manual work by an analyst.

How Agentic AI Thinks & Works

PATTERNS OVERVIEW

6 STRATEGIES THAT POWER AGENTIC AI — EACH SOLVES A DIFFERENT KIND OF PROBLEM

Agentic AI doesn't use just one fixed method — it applies different "thinking strategies" depending on the task. These are called Agentic Patterns. There are many emerging patterns in the research community, but the 6 shown here are the most widely adopted and proven in real-world systems today. Think of them as the essential problem-solving playbooks every AI builder should know.

01



ReAct

Think, then act, then observe — and repeat. Like a detective who forms a theory, checks a clue, updates the theory, checks again, until the case is solved.

04



Tree of Thought

Explore several different approaches at once, score them, and pick the best. Like a chess player who mentally plays out multiple moves before choosing one.

02



Plan & Execute

Make a full to-do list first, then work through it step-by-step. Best when the task has a clear sequence of actions — like building a house from a blueprint.

05



Graph Workflows

Map the task as a flowchart with branches and decision points. Perfect for complex business processes where different paths lead to different outcomes.

03



Reflection

Do the work, then review your own output and improve it. Like a writer who drafts an essay, reads it critically, and rewrites the weak parts before submitting.

06



Hierarchical Agents

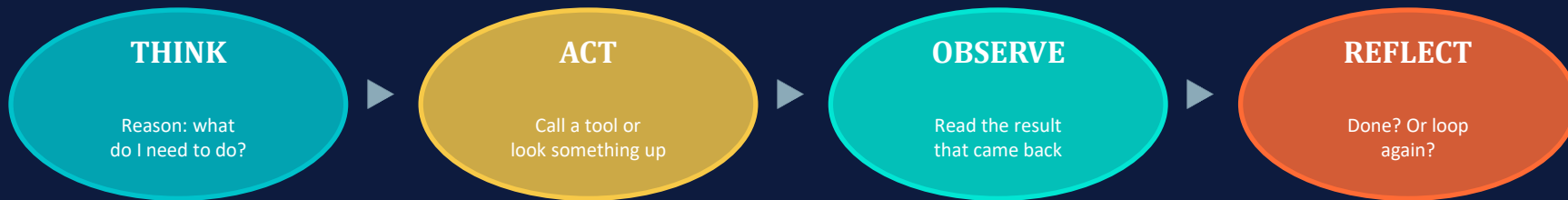
A manager AI delegates sub-tasks to specialist AIs. Like a project manager who assigns research to one person, writing to another, and review to a third.

Pattern 1 — ReAct Strategy

AGENTIC PATTERN

THINK FIRST, ACT NEXT, LEARN FROM WHAT HAPPENS — THEN REPEAT

ReAct stands for Reasoning + Acting. Instead of guessing the answer right away, the AI thinks about what it needs to do, takes one small action (like searching for information), looks at the result, then thinks again about the next step. This loop continues until the task is done. It's like a detective who forms a theory, checks a clue, updates their theory, and keeps going until the case is solved.



↺ Loops until the answer is found or a step limit is reached



Examples — How ReAct Works in Practice

Customer Service Bot: THINK → "What's the status of this customer's return?" | ACT → Search order database | OBSERVE → Order #4521 shipped 3 days ago | THINK → "Is it late?" | ACT → Check expected delivery date | OBSERVE → Expected tomorrow | REFLECT → Reply: "Your order arrives tomorrow."

Medical Query Assistant: THINK → "Member asks if their MRI is covered." | ACT → Look up their insurance plan | OBSERVE → Gold PPO plan, outpatient imaging covered | ACT → Check if prior authorization is needed | OBSERVE → PA required for MRI | REFLECT → Reply with coverage details and PA instructions.

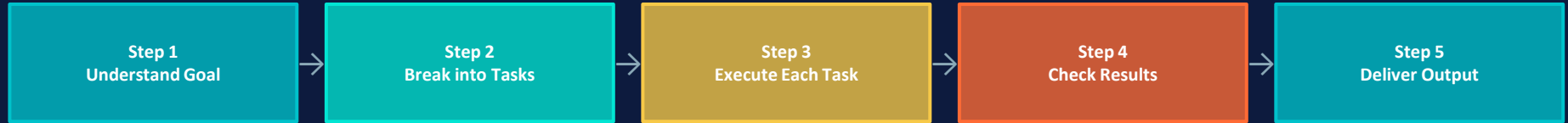
Research Helper: THINK → "Find recent AI regulations in the EU." | ACT → Web search | OBSERVE → EU AI Act passed 2024 | ACT → Fetch full document summary | REFLECT → Write concise briefing for the reader.

Pattern 2 — Plan & Execute

AGENTIC PATTERN

MAKE A FULL PLAN FIRST, THEN CARRY IT OUT STEP BY STEP — GREAT FOR PREDICTABLE WORKFLOWS

Plan & Execute splits the job in two: a Planner AI designs the full list of steps needed to complete the goal, then an Executor AI carries each step out in order. It's like a contractor who draws up a blueprint before picking up any tools. This approach is more reliable for complex tasks because the system knows the whole plan upfront — and can explain every decision it makes.



Trip Planning

Goal: "Book a 5-day Paris trip for 2, budget \$4,000."

Plan: [1] Search flights → [2] Search hotels → [3] Check availability → [4] Compare costs → [5] Book best option → [6] Email confirmation.

Each step is handled by a specialized tool. The planner monitors and adjusts if a step fails — e.g., if the preferred hotel is full, it automatically tries the next best.



Monthly Business Report

Goal: "Produce the Q3 performance report."

Plan: [1] Pull sales data → [2] Calculate metrics → [3] Spot any unusual trends → [4] Write summary text → [5] Create charts → [6] Format as PDF.

This replaces a process that used to take a 3-person team 2 days — now can run automatically overnight.



Insurance Claims Processing

Goal: "Adjudicate this medical claim."

Plan: [1] Verify member is eligible → [2] Check what the procedure code covers → [3] Apply deductible → [4] Calculate patient's share → [5] Flag anything unusual for human review → [6] Send Explanation of Benefits.

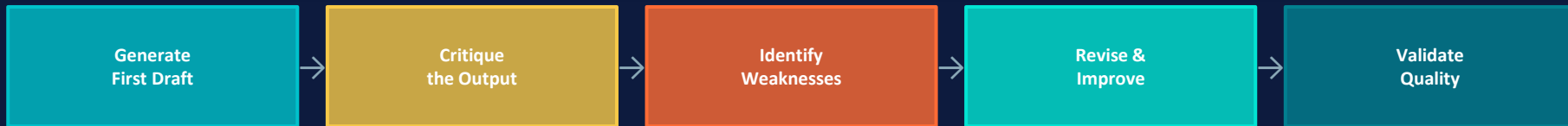
Can significantly reduce manual adjudication effort and accelerate claims processing in production deployments.

Pattern 3 — Reflection

AGENTIC PATTERN

AI CHECKS ITS OWN WORK AND IMPROVES IT — LIKE A WRITER WHO EDITS BEFORE SUBMITTING

Reflection is the AI's ability to review, critique, and improve its own output. After generating an answer or document, a second AI review pass (or the same AI looking at it fresh) checks for errors, gaps, or quality issues — and revises the work. This loop continues until the output meets a quality bar. It dramatically reduces mistakes and hallucinations (made-up facts) in AI-generated content.



↺ Loops until quality is acceptable — then outputs the final result

Code Writing & Self-Correction

AI writes a piece of software, then reviews it looking for bugs, security holes, and edge cases it missed. It rewrites the weak parts and runs automated tests. Reflection and self-review techniques have been shown to improve code quality and reduce errors compared to single-pass generation. Claude Code uses this pattern by default.



Medical & Legal Document Review

AI drafts a prior authorization letter. A second review pass checks: Are the diagnosis codes correct? Is the required legal language included? Are any mandatory fields missing? The AI corrects all issues before the document goes to a human for signature — catching errors that manual review often misses.



Report Accuracy Checking

AI generates a financial or performance summary. The reflection pass verifies every number against the source data, flags any figure that differs by more than 5%, and adds proper citations. This can dramatically reduce manual quality-check effort while improving consistency.

Pattern 4 — Tree of Thought

AGENTIC PATTERN

EXPLORE MULTIPLE APPROACHES SIMULTANEOUSLY — SCORE THEM — PICK THE BEST



What is Tree of Thought?

Instead of picking one approach right away, the AI thinks like a chess player — it imagines several possible paths forward at once, scores each one, prunes the weak ones, and digs deeper into the best option. Perfect for complex decisions with no single obvious answer.

Define Goal

Path A

Weak

Path B ★

Best!

Path C

Weak

★ Best path selected and fully executed as final answer

Best for: Strategy decisions, architectural choices, diagnosis, creative problem-solving

Tree of Thought in Action



Choosing a Market Entry Strategy

A company evaluates entering a new market. The AI explores 4 paths in parallel: build a local office, acquire a startup, form a distributor partnership, or license its technology. Each path is scored on cost, speed, and revenue potential. The best-scoring option — partnership — is presented with a full business rationale, accelerating strategic analysis and scenario evaluation..



Financial Portfolio Optimization

Given an investor's goals, the AI simultaneously models 5 portfolio configurations — conservative bonds, growth equities, balanced mix, sector-focused, and ESG-tilted. Each is stress-tested against historical scenarios. The highest risk-adjusted configuration is recommended with a plain-English explanation of every tradeoff.



Narrowing a Medical Diagnosis

A patient has symptoms matching 6 possible conditions. The AI explores all 6 simultaneously, scores each against lab results and clinical guidelines, prunes the 4 least likely, and surfaces the top 2 with recommended confirmatory tests. The physician reviews the AI's scored shortlist and makes the final clinical decision.

Pattern 5 — Graph-Based Workflows

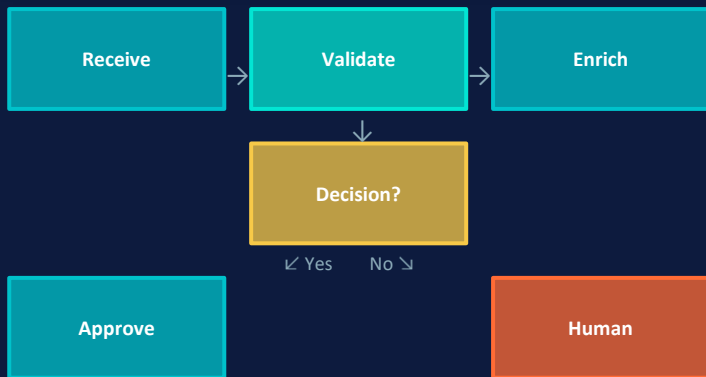
AGENTIC PATTERN

MAP THE WHOLE PROCESS AS A FLOWCHART — EACH BOX IS AN AI STEP



What are Graph Workflows?

A graph workflow maps an entire process as a flowchart. Each box is an AI step — check eligibility, generate a document, route to a human. The arrows represent what happens next based on the result: approved → auto-process; unclear → human review. Frameworks like LangGraph are built for this exact pattern.



★ Human approval gates can be inserted anywhere — critical for regulated industries

Graph Workflows in Action



Loan Approval Pipeline

A bank uses a graph to automate loan decisions. Nodes: Receive application → Verify identity → Pull credit score → Check income → Calculate risk → Auto-approve if criteria met → Route to underwriter if borderline → Notify applicant. Human underwriters focus only on genuinely ambiguous cases, improving efficiency and consistency.



Employee Onboarding Workflow

HR uses a graph for new hire onboarding. Nodes: Offer accepted → Create accounts → Send welcome pack → Assign buddy → Schedule Day 1 orientation → Provision laptop → Enroll in benefits. If any step fails (e.g., IT can't provision in time), the graph alerts the right team automatically — ensuring no new hire has a bad first day.



Content Moderation System

A social platform uses a graph to review user posts. Nodes: Post submitted → AI confidence check → If clearly safe: approve instantly → If clearly harmful: remove → If ambiguous: route to human moderator → Log decision. Most routine cases can be automated, allowing human moderators to focus on complex edge cases.

Pattern 6 — Hierarchical Agents

AGENTIC PATTERN

A MANAGER AI ASSIGNS WORK TO SPECIALIST AIs — JUST LIKE A HUMAN TEAM



What are Hierarchical Agents?

Think of a project manager who receives a big goal, breaks it into tasks, and assigns each task to the specialist best suited for it. A Manager AI does the same: it delegates sub-tasks to Specialist AIs — a researcher, a writer, a coder — monitors their outputs, and assembles the final deliverable. Complex projects handled like a coordinated team.



Best for: Long multi-step projects that benefit from specialized expertise per sub-task

Hierarchical Agents in Action



Competitive Intelligence Report

Manager receives: "Research our top 3 competitors." Delegates: Researcher Agent scrapes websites and news, Financial Agent pulls public filings, Product Agent compares feature lists, Writer Agent drafts the narrative, Reviewer Agent fact-checks every claim. Tasks that traditionally required days can often be completed in hours through coordinated agent workflows.



Multi-Language Content Campaign

Marketing goal: "Launch our product in 5 countries." Manager delegates: Copywriter Agent writes English master content, then separate Localizer Agents for Spanish, French, German, and Japanese each adapt tone and cultural references. SEO Agent optimizes for each market. Compliance Agent checks regulations. Full 5-market campaign in 2 hours.



IT System Incident Response

A production outage occurs at 2am. Manager AI activates: Diagnostic Agent identifies root cause, Rollback Agent reverts the last deployment, Monitor Agent watches system metrics, Communication Agent drafts the incident report, Postmortem Agent generates action items. Engineers wake up to a resolved incident with a full timeline — instead of a 4am emergency call.

Multi-Agent Systems

ADVANCED · TEAMWORK

MULTIPLE AI AGENTS WORK AS TEAMMATES — EACH HANDLES A PIECE, THEN SHARES RESULTS



What is a Multi-Agent System?

A Multi-Agent System (MAS) is a group of AI agents working as teammates. Unlike Hierarchical Agents where one is the boss, in MAS the agents are often peers — each working on its own piece of the problem, then sharing results.

Think of it like a task force: a group of specialists who each contribute their expertise independently, then combine findings at the end.

Connection to Agentic AI: Multi-agent systems are the most advanced form of Agentic AI — instead of one AI doing everything, a coordinated team of AIs divides and conquers.

How MAS relates to Agentic AI

- ▶ **Single Agent:** One AI, one goal, one chain of steps
- ▶ **Agentic AI:** One orchestrator + multiple tools/sub-agents
- ▶ **Multi-Agent:** Peer agents collaborate; no single controller required

Multi-Agent Systems in Action



CrewAI — Research & Writing Crew

A crew of 5 agents works on a competitive analysis: Lead Researcher finds market data, Web Scraper pulls competitor pricing, Fact Checker verifies each claim, Analyst finds the patterns, Writer produces the final report. Parallel agent collaboration can dramatically accelerate research and synthesis tasks.



AutoGen — Debate & Peer Review

Two agents — a Proposer and a Critic — debate a proposed solution (e.g., a new business process). The Proposer argues for it with evidence; the Critic challenges every claim. After 3–5 rounds, they reach consensus on the strongest version of the idea. This mimics expert peer review at machine speed.



LangGraph Swarm — Complex Case Handling

A customer service case is handled by a swarm: Claims Agent checks history, Eligibility Agent verifies coverage, Billing Agent calculates balance, Communication Agent drafts the reply. All work in parallel on the same case, share findings through a shared memory, and the result is assembled automatically — dramatically reducing case handling times..

Physical AI — Robots & the Real World

FRONTIER · NOW

AI THAT MOVES, SEES, AND ACTS IN THE PHYSICAL WORLD — NOT JUST ON A SCREEN

Physical AI is what happens when AI intelligence is given a body. Instead of just answering questions, the AI can see through cameras, hear through microphones, and move things with robotic arms or legs. These systems combine the reasoning power of large language models with sensors and actuators to interact with the physical world — in factories, hospitals, homes, and on roads.



Humanoid Robots

Robots shaped like humans, designed to work alongside people.

- **Tesla Optimus:** Works on Tesla's own factory floor, sorting parts, moving boxes. Uses the same AI vision system as Tesla cars. Gen 2 works 40% faster than Gen 1.
- **Figure AI:** Backed by OpenAI. Understands spoken instructions ("load the dishwasher"), sees the environment, and carries out the task. Now deployed in BMW factories for precision assembly.



Industrial Automation Robots

AI-powered machines that handle manufacturing, warehousing, and logistics.

- **Amazon Robotics:** Thousands of AI robots navigate warehouses autonomously, picking and packing orders 3x faster than humans — Robotics now play a major role in Amazon's fulfillment operations.
- **Boston Dynamics Spot:** A four-legged robot used in construction sites and oil rigs to inspect equipment, detect leaks, and create 3D maps — in environments too dangerous for humans.



Autonomous Vehicles

Self-driving cars and delivery robots that navigate real roads.

- **Waymo (Google):** Fully driverless taxis operating in Phoenix, SF, and LA — over 1 million paid rides with zero fatalities. The AI sees 360° with cameras, radar, and lidar.
- **Nuro:** AI delivery vehicles operating in grocery and pharmacy delivery — navigating neighborhoods without any human onboard.



Healthcare Robots

Robots assisting in surgery, patient care, and hospital logistics.

- **Da Vinci Surgical System:** AI-assisted robot enables surgeons to perform minimally invasive procedures with precision beyond human hand capability — used in 10+ million procedures worldwide.
- **TUG Robots (Aethon):** Autonomous hospital robots delivering medications, linens, and supplies through hospital corridors 24/7 — Helping hospitals improve reliability and reduce operational errors.

Artificial General Intelligence (AGI)

FUTURE · NOT YET HERE

AI THAT CAN DO ANYTHING A HUMAN CAN DO — ACROSS EVERY SUBJECT AND SKILL



What is AGI?

Today's AI is excellent at specific tasks — writing, coding, image recognition. But it can't learn a new skill on its own or apply knowledge from one area to a completely different one.

AGI is the next leap: an AI that's a generalist. It could pass any professional exam, learn a new language or skill from scratch, do original scientific research, and understand social and emotional context — just like a human expert can across many domains.

Imagine one AI that could be your doctor, lawyer, financial advisor, and research scientist simultaneously — and could keep learning and improving on its own.

⚠️ Status: Not Yet Achieved (June 2026)

Today's best AI models (Claude, GPT-5, Gemini) are extremely capable but still narrow in important ways. They can fail at physical common sense, sustained new learning, and understanding context the way humans naturally do.

Several AI labs believe AGI could emerge within the coming decade, though timelines remain uncertain.

What AGI Could Enable



Accelerating Medical Research

An AGI system could simultaneously read every published medical study, model how diseases progress, design new drug candidates, simulate how they'll interact with human biology, and propose clinical trial designs — compressing decades of research into years. Alzheimer's, cancer, and rare diseases could have cures within a decade.



Personalized Education for Everyone

An AGI tutor could understand exactly how each student learns, adapt teaching methods in real time, identify knowledge gaps, and provide the equivalent of a world-class private tutor to every child on Earth — regardless of their economic background or geography.



Climate & Science Problem Solving

AGI could simultaneously model climate systems, design new materials for solar cells and batteries, optimize global energy grids, and propose policy interventions — tackling problems too complex for any human team to solve in a reasonable timeframe.



Enterprise Readiness Implication

Organizations that build strong AI governance, data quality, and human-oversight frameworks TODAY will be positioned to safely adopt AGI when it arrives. Those without these foundations face profound risks. The time to prepare is now — not when AGI is at the door.

Artificial Superintelligence (ASI)

THEORETICAL FUTURE

AI THAT IS SMARTER THAN THE SMARTEST HUMANS IN EVERY WAY — A THEORETICAL BUT IMPORTANT CONCEPT

ASI is the concept of an AI that exceeds the best human experts in every domain simultaneously — science, creativity, strategy, social understanding, and more. It could think faster, learn faster, and solve problems no human mind could solve. This does not exist today and may not for decades — but understanding it matters because the decisions we make today about AI safety will determine how this plays out.



Drug Discovery at Impossible Speed

Today it takes 10–15 years and \$2B+ to bring a new drug to market. An ASI could simultaneously model every known protein, design millions of candidate molecules, predict their safety profile, and simulate clinical trial outcomes — potentially discovering cures for diseases like Alzheimer's or could potentially accelerate.

Real signal today: AlphaFold (DeepMind) already solved protein structure prediction in 2021 — a problem biologists worked on for 50 years.



Self-Improvement (some researchers speculate)

The most consequential ASI property: it could rewrite and improve its own code, making itself smarter. A smarter version then improves itself further — creating a rapidly accelerating cycle. Futurist Ray Kurzweil called this the "Singularity."

This is why AI safety researchers like those at Anthropic are working today on making AI systems controllable and aligned with human values — before this capability arrives.



The Alignment Challenge

The central question of ASI is: how do we ensure a system far smarter than us does what we actually want? If an ASI's goal is subtly misaligned with human values — even unintentionally — the consequences at that level of capability could be severe.

Anthropic's mission is specifically to solve this problem. Constitutional AI, interpretability research, and careful deployment policies are today's building blocks for a safe ASI future.



What Organizations Should Do Today

ASI may still be far away — but the habits and infrastructure you build now matter enormously:

- Implement AI governance and human oversight now, not later
- Invest in data quality and model interpretability
- Train employees to work alongside AI systems
- Follow safety research from Anthropic, DeepMind, and OpenAI
- Build a culture of responsible AI use from the ground up

Challenges & Guardrails

RESPONSIBLE AI

AS AI BECOMES MORE POWERFUL, USING IT RESPONSIBLY BECOMES MORE IMPORTANT



Safety & Alignment

- AI systems must be designed to pursue the goals we actually intend — not a subtly wrong version of them. This is called "alignment."
- Output guardrails — filters, confidence checks, human review gates — prevent AI errors from reaching the end user directly.
- Human-in-the-loop checkpoints are critical in high-stakes decisions (medical, legal, financial) where errors have real consequences.
- Responsible scaling: Anthropic links AI capability releases to passing verified safety evaluations before public deployment.



Privacy, Security & Data

- Sensitive data (health records, financial info, personal data) must never enter AI systems without proper consent, contracts, and security controls.
- AI systems can be manipulated through "prompt injection" — bad actors hiding instructions in content the AI reads. Defenses are required.
- Access controls must limit which AI agents can access which data — not every AI should see everything.
- Regulatory requirements (HIPAA, GDPR, SOC2) apply to AI systems just as they do to any other enterprise software.



Human Oversight & Ethics

- AI should augment human judgment, not replace it — especially for decisions that affect people's lives, livelihoods, or wellbeing.
- Organizations need clear policies on where AI can act autonomously vs. where a human must review and approve.
- AI systems should be explainable: people affected by AI decisions have a right to understand why the AI made that decision.
- Bias in training data leads to biased AI outcomes. Regular audits of AI decisions are necessary to catch and correct this.

Key Principle: Governance, safety, and ethics are not obstacles to AI adoption — they are what makes AI adoption sustainable and trustworthy.

MCP — Model Context Protocol

AGENTIC AI ENABLER

THE UNIVERSAL PLUG THAT LETS AI AGENTS CONNECT TO ANY TOOL OR SERVICE



What is MCP?

MCP (Model Context Protocol) is an open standard created by Anthropic that gives AI agents a consistent, universal way to connect to any external tool, database, or service.

Think of it like a USB port for AI. Without MCP, every AI needs custom code to connect to every tool separately. With MCP, one universal plug works for everything — swap tools without rewriting the AI.

Without MCP: custom integration code per tool, per AI — expensive and brittle.
With MCP: one standard. Any compliant AI calls any compliant tool instantly.

How it works — 3 steps:

1. AI Agent sends a request in MCP standard format
2. MCP Server receives it and calls the real tool or API
3. Result is returned to the agent in a standard format

Why MCP Matters for Agentic AI



MCP = the "translator" layer between AI brain and any external capability



AI Agent + Web Search

Without MCP: developer writes custom code to connect AI to Google, Bing, DuckDuckGo separately.
With MCP: agent calls one MCP-compliant search server. Switch providers without rewriting the agent.
Claude.ai already uses MCP to power its web search tools today.



AI Agent + Enterprise Systems (Salesforce, SAP, ServiceNow)

A company has 3 business systems. With MCP, one AI agent calls all three with the same interface. Adding a new system means connecting it to MCP once — the agent automatically gains access. No bespoke integration per system, per AI.



AI Agent + Secure Internal APIs (MCP Gateway)

An org builds an MCP Gateway in front of internal APIs (billing, eligibility, claims). Any approved agent calls these safely through MCP — with access control, rate limiting, and audit logging enforced at the gateway. No agent ever touches backend systems directly.

MCP in Action — The Agentic AI Connection

MCP · ARCHITECTURE

MCP IS THE INFRASTRUCTURE LAYER THAT MAKES AGENTIC AI PRACTICAL AT ENTERPRISE SCALE

Agentic AI is only as powerful as the tools it can reach. MCP solves the "last mile" problem: how does an AI agent safely and reliably call a company's real systems — CRM, databases, identity services — without custom code for every connection? MCP creates a governed, auditable, standardized interface between every AI agent and every tool it needs to do real work.

AI Agent (Claude / GPT / LangGraph / Custom)

MCP Client (built into the agent framework)

MCP Gateway (access control · logging · routing)

Backend Tools & Services (APIs · DBs · search · code runners)

Each layer is decoupled — swap any AI, gateway, or backend without rebuilding the system



Customer Service Agent via MCP

User: "What's the status of my claim #47821?"
→ Agent calls ClaimsStatusTool via MCP → Gateway authenticates & routes → Claims API returns data → Agent replies in natural language.
The agent never directly touches the claims database. Every call is logged for audit.



Developer Coding Agent via MCP

Developer: "Run the tests and fix any failures."
→ Agent calls CodeRunnerTool (MCP) → runs tests in sandbox → returns failures → Agent calls FileTool (MCP) → edits code → re-runs → all pass.
MCP ensures AI only accesses whitelisted directories; no arbitrary system access.



Research Agent — 3 Tools via MCP

Researcher: "Summarize new GLP-1 drug trials this month."
→ Agent calls WebSearchTool (MCP) → reads 12 articles → DocReaderTool (MCP) → extracts findings → CitationTool (MCP) → formats references → delivers structured briefing.
All 3 tool calls go through MCP with independent access control.

RAG & Agentic RAG

KNOWLEDGE RETRIEVAL

GIVING AI ACCESS TO YOUR OWN DOCUMENTS — SO IT ANSWERS FROM YOUR DATA, NOT JUST ITS TRAINING

RAG (Retrieval-Augmented Generation) lets an AI look up information from a document library before answering — like giving it access to your company's internal knowledge base. Without RAG, AI only knows its training data. With RAG, it searches your policies, reports, and databases in real time. Agentic RAG goes further: the agent decides when to search, what to search for, validates what it finds, and searches again if needed.



1 • Basic RAG

Ask → search docs → add results to prompt → answer.

e.g. "What is our return policy?" → AI searches policy PDF → answers directly from the document.



2 • Iterative RAG

Search → read → if answer incomplete → search again with better query.

e.g. Complex legal question → first search gives partial answer → AI refines query → searches again → combines both results.



3 • Self-RAG

AI first decides: do I even need to search, or do I already know this?

e.g. "What's 2+2?" → AI skips retrieval. "What's our Q3 revenue?" → AI retrieves the financial report first.



4 • Corrective RAG (CRAG)

After retrieving, AI grades each document — discards irrelevant ones, re-searches if needed.

e.g. Query returns 5 docs. AI scores each. 3 are off-topic, discarded. AI re-searches with a sharper query before answering.



5 • Graph RAG

Documents stored as a knowledge graph — AI traverses relationships, not just keyword matches.

e.g. "Who approved this account and which policies apply?" → AI traverses Person → Account → Approvals → Policies in the graph.



6 • Agentic RAG

A full AI agent controls the entire retrieval strategy autonomously — no human directing each step.

e.g. "Summarize competitor pricing trends" → agent searches web + internal reports → cross-validates → flags conflicts → writes cited brief.

Agent Gateway & Agentic Harness

AGENTIC AI INFRASTRUCTURE

THE CONTROL LAYER THAT GOVERNS WHAT AI AGENTS CAN SEE, DO, AND CALL

An Agent Gateway (also called an Agentic Harness) is the infrastructure layer that sits between AI agents and everything they interact with. It controls which agents can call which tools, enforces rate limits and cost budgets, logs every action for audit, and routes requests to the right backend services — all transparently.

Simple analogy: MCP = the plug standard (how agents talk to tools). Agent Gateway = the power board (what gets plugged in, who is authorized, and how much power each device can draw).



What an Agent Gateway Does



Access Control

Decides which agents can call which tools or APIs. A customer-service bot should never touch the billing override endpoint — the gateway enforces that rule automatically.



Rate Limiting & Cost Control

Prevents runaway agents or infinite loops from making thousands of API calls per minute. Keeps cloud costs predictable and protects backend services from overload.



Audit Logging

Records every tool call, every input, every output with timestamps. You can replay exactly what any AI agent did and when — essential for compliance and debugging.



Routing & Failover

Directs agent requests to the right service, handles failover when a service is down, and can dynamically route between AI model providers based on cost or latency.

Where the Gateway Fits — Architecture View

AI Agents (Claude · GPT · Gemini · LangGraph)

↓ *every request must pass through* ↓

Agent Gateway / Harness (Auth · Rate Limit · Route · Log)

↓ *only approved, rate-limited calls continue* ↓

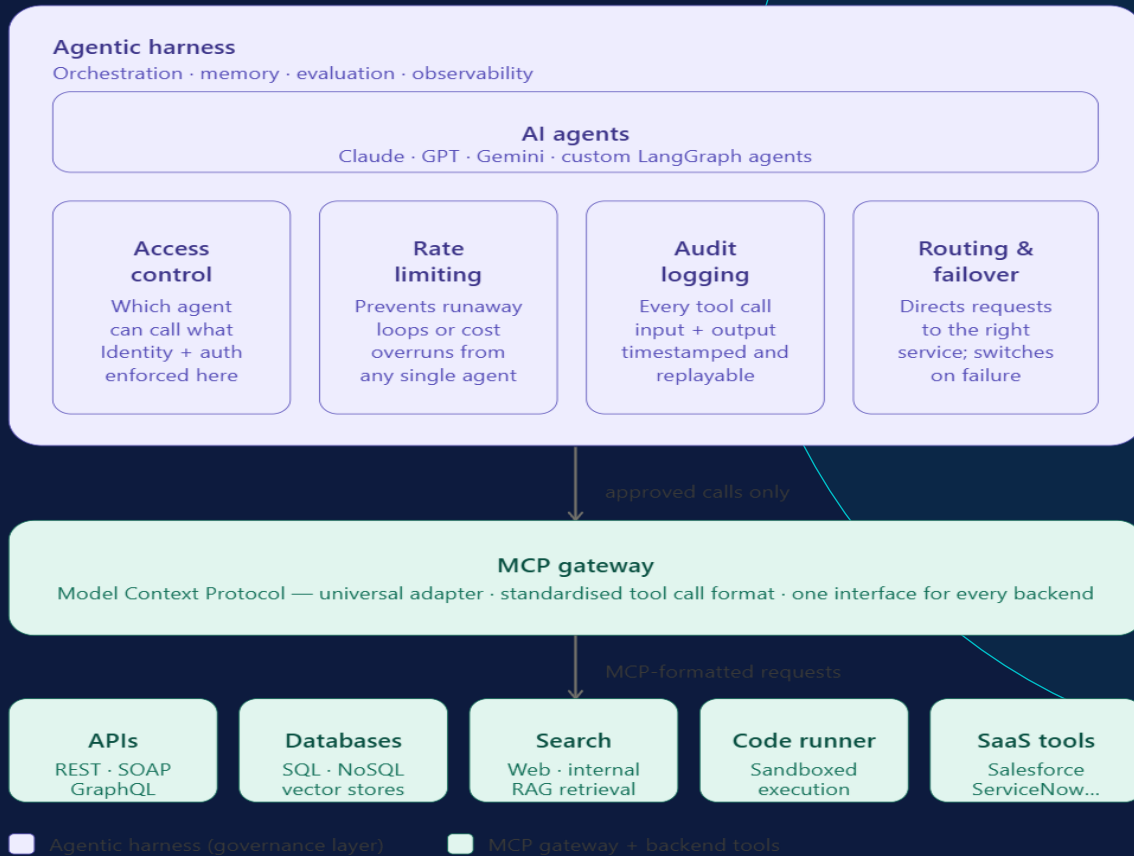
MCP Layer (standardized tool call protocol)

↓

Backend Services (APIs · Databases · Tools)

★ The Gateway is the safety net — agents can be ambitious, but the gateway enforces the rules.

MCP and Agentic Harness



Harness = the rules engine · MCP = the universal plug

KEY TAKEAWAYS

The Road Ahead



AI Agents are here NOW

Deployed across customer service, coding, research, and writing. This is the baseline — not the future.



Agentic AI = full workflow automation

6 key patterns power this: ReAct, Plan & Execute, Reflection, Tree of Thought, Graph Workflows, and Hierarchical Agents.



MCP connects agents to tools

The open standard that lets any AI agent talk to any tool safely — the plug that makes interoperability real.



Agent Gateways enforce the rules

AWS AgentCore, Azure AI Foundry, Google Vertex, LangGraph + Kong — all govern agents at scale with audit, access control, and rate limiting.



RAG + Governance = trusted AI

RAG grounds AI in your real data. Human oversight, guardrails, and audit logs keep it accountable. Both are required for safe enterprise deployment.

AI Skills & Workforce Readiness

ROLES IN DEMAND · SKILLS TO SCREEN FOR · UPSKILLING ROADMAP · LEADERSHIP



Roles to hire immediately

- Agentic AI Engineer — LangChain, LangGraph, MCP, multi-agent orchestration
- Prompt Engineer / AI Interaction Designer — evaluation, guardrail & HITL design
- AI / MLOps Infrastructure Engineer — model deployment, token cost, latency tuning
- AI Solutions Architect — translates business processes into agentic workflow designs
- AI Safety & Governance Analyst — bias audits, regulatory compliance (EU AI Act, SOC2)



Baseline AI fluency — screen every new hire

- Can explain the difference between an AI Agent and Agentic AI without jargon
- Hands-on experience with at least one LLM API (Anthropic, OpenAI, or Gemini)
- Understands RAG — retrieval, chunking, embedding, and grounding in plain language
- Familiar with MCP or equivalent tool-calling / agent gateway standards
- Aware of AI governance requirements relevant to your clients' industries

Upskilling roadmap — existing staff



0 – 6 months · Foundation

Python for AI, LangChain fundamentals, @tool decorator · Prompt engineering and evaluation basics · RAG pipeline — retrieval, chunking, grounding · Call a real LLM API; build a working agent



6 – 12 months · Agentic

LangGraph stateful graphs, checkpointing, HITL · MCP server/client design and agent gateway patterns · Multi-agent orchestration (CrewAI, AutoGen, Send()) · Agent harness platforms — AWS, Azure, Vertex AI



12 – 18 months · Leadership

AI governance, safety, and EU AI Act compliance · AGI transition planning and workforce strategy · Physical AI awareness — robotics, autonomous systems · AI practice P&L ownership and pricing models

Key insight: scarcest resource = people who understand BOTH business process AND agentic technology. Hire for that intersection first.

Leadership actions: Build AI CoE · Draft AI governance charter · Create sandboxes · Partner with cloud vendor for credits & training · Price services on AI leverage, not just headcount

References & Further Reading

FOR ATTENDEES

OFFICIAL DOCS · RESEARCH PAPERS · PLATFORMS · SAFETY & GOVERNANCE



Anthropic & MCP

Docs	docs.anthropic.com — tool use, Claude models, API strings
Free	anthropic.com/claude/fable — Claude Fable 5 overview
Free	anthropic.com/news — Claude Fable 5 & Mythos 5 launch post
Free	anthropic.com/glasswing — Project Glasswing / Claude Mythos
Docs	modelcontextprotocol.io — MCP spec & server registry
Free	anthropic.com/research/building-effective-agents



Cloud Agent Platforms

Docs	AWS Bedrock AgentCore — aws.amazon.com/bedrock/agentcore
Docs	Azure AI Foundry — azure.microsoft.com/products/ai-foundry
Docs	Google Vertex AI Agent Builder — cloud.google.com/agent-builder
Docs	IBM watsonx.ai + Governance — ibm.com/products/watsonx-ai
Docs	Kong Enterprise MCP Gateway — konghq.com/products/kong-gateway
Docs	WSO2 API Manager (MCP adapter) — wso2.com/api-manager



Agentic Patterns & Frameworks

Paper	ReAct: Reasoning + Acting — arxiv.org/abs/2210.03629
Paper	Tree of Thought — arxiv.org/abs/2305.10601
Paper	Reflexion (Reflection pattern) — arxiv.org/abs/2303.11366
Free	CrewAI multi-agent framework — crewai.com
Free	AutoGen (Microsoft) — microsoft.github.io/autogen
Free	github.com/anthropics/anthropic-sdk-python



LangChain & LangGraph

Docs	python.langchain.com/docs — official LangChain documentation
Docs	langchain-ai.github.io/langgraph — LangGraph official docs
Docs	LangGraph: agentic concepts, multi-agent, Send(), HITL langchain-ai.github.io/langgraph/concepts/persistence — checkpointing blog.langchain.dev — patterns, tutorials, agent examples
Free	github.com/langchain-ai/langgraph — source & notebooks



RAG & Retrieval

Docs	LangChain RAG concepts — python.langchain.com/docs/concepts/rag
Paper	Corrective RAG (CRAG) — arxiv.org/abs/2312.10997
Paper	Self-RAG — arxiv.org/abs/2310.11511
Paper	Graph RAG (Microsoft) — arxiv.org/abs/2404.16130
Free	github.com/microsoft/graphrag — Graph RAG open source
Docs	Vector stores in LangChain — langchain.com/docs/vectorstores



Physical AI, AGI & Safety

Docs	NVIDIA Isaac — Physical AI & robotics — developer.nvidia.com/isaac
Free	Figure AI humanoid robot — figure.ai
Free	AlphaFold (AGI signal) — deepmind.google/alphafold
Free	Anthropic responsible scaling policy — anthropic.com/rsp
Free	EU AI Act full text — artificialintelligenceact.eu
Free	OpenAI Superalignment — openai.com/research/superalignment

Free = publicly accessible

Docs = official documentation
↑ All entries above are live hyperlinks — click any line to open in your browser

Paper = peer-reviewed research

Appendix

Technical Reference Material:

- Capability Matrices
- Platform Comparisons
- Agentic Harness Products

Agentic Harness products capability Matrix

CAPABILITY MATRIX — AT A GLANCE

Capability	AWS AgentCore	Azure AI Foundry	Google Vertex	Anthropic	LangGraph+Kong	IBM watsonx
Harness + Gateway unified	Yes — single product	Two products	Two products	Delegated	Self-assembled	Two products
Managed memory store	Built-in	Cosmos + AI Search	Feature Store	Bring your own	Bring your own	Db2 + Discovery
Built-in evaluation	AgentCore eval hooks	Azure AI Eval	Agent Eval Service	Via LangSmith	LangSmith add-on	Governance overlay
Native MCP support	MCP gateway native	APIM adapter 2025	Apigee adapter 2025	Protocol inventor	Kong adapter 2025	Partner ecosystem
LangGraph hosting	Native support	Via container	Via Agent Engine	Indirect	Self-hosted	Limited
No-code agent builder	Partial	Copilot Studio	Agent Builder	Not available	Not available	Partial
Compliance / governance	IAM + CloudTrail	Entra + Monitor	VPC + Cloud IAM	Enterprise plan	Kong policies	Deepest in market
On-premises option	Cloud only	Arc hybrid	Anthos hybrid	Cloud only	Fully on-prem	On-prem strong

Yes = strong native capability Partial = available but requires assembly or add-on

Agentic Harness Products(AWS AgentCore)

AWS BEDROCK AGENTCORE — THE ONLY UNIFIED HARNESS + GATEWAY PRODUCT

AWS Bedrock AgentCore

Harness + MCP Gateway unified

The only major platform that ships both halves — harness and MCP gateway — as a single integrated product

Harness side

Managed memory store (short + long-term). Secure code execution sandbox. IAM-enforced identity for agents. Built-in eval hooks. Auto-scaling runtime.

MCP Gateway side

AgentCore Gateway routes MCP-formatted tool calls to AWS services (Lambda, S3, DynamoDB, SageMaker) with IAM policy enforcement per tool. Native MCP server registry.

Why it's unified

AWS designed it so the harness and gateway share the same IAM identity layer. An agent's permissions in the harness automatically propagate to its MCP tool call rights — no separate config.

Best for teams already in AWS. Supports Claude, Titan, Llama, Mistral as the underlying model. LangGraph agents can be hosted directly inside AgentCore.

Agentic Harness products(Azure and Google Cloud)

OTHER MAJOR VENDOR HARNESSES AND GATEWAYS

Microsoft Azure

AI Foundry Agent Service + Azure API Management

Harness	Azure AI Foundry manages agent lifecycle, tool registration, Entra ID auth, and Azure Monitor observability. Semantic Kernel is the orchestration SDK underneath. Strong
MCP Gateway	Azure API Management serves as the gateway layer — it's a mature enterprise API gateway but was not MCP-native originally; MCP adapter support added in 2025. Partial
Memory	Azure Cosmos DB and Azure AI Search as managed long-term stores. Short-term via Semantic Kernel context.
Unique strength	Copilot Studio — no-code harness layer. Deep M365/Graph/Teams/Dynamics integration. Best for Microsoft-native. •
Gap vs AWS	Harness and API gateway are two separate products you configure together — not unified like AgentCore.

Google Cloud

Vertex AI Agent Builder + Agent Engine + Apigee

Harness	Agent Engine is the managed agent runtime. Agent Builder is the no-code builder on top. Vertex AI Extensions = tool registry. Agent Eval Service built in. Strong
MCP Gateway	Apigee API Gateway handles routing, auth, rate limiting. Google added MCP server support in 2025. Apigee is separate from Agent Builder. Partial
Memory	Vertex AI Feature Store, BigQuery, AlloyDB for long-term. In-context for short-term. Grounding with Google Search natively built in.
Unique strength	Best real-time grounding story — agents can natively search Google and retrieve live data. ADK (Agent Development Kit) for rapid prototyping.
Gap vs AWS	Agent Engine and Apigee are separate products; unified identity layer between them requires manual IAM wiring.

Agentic Harness products (Anthropic & Open Source)

Anthropic

Claude Agent SDK + MCP (protocol) + Enterprise

Harness	No managed harness of its own. Claude Agent SDK provides primitives (tool use, multi-turn, memory hooks). Harness is delegated to AWS AgentCore, Azure, or self-built. Delegated
MCP Gateway	Anthropic invented and owns the MCP standard. Every MCP-compatible gateway and server speaks the protocol Anthropic defined. Huge ecosystem leverage. Protocol owner
Memory	No managed memory store. You bring your own (Pinecone, pgvector, Redis) and wire it via MCP tool or LangChain memory wrapper.
Unique strength	By owning MCP, Anthropic's model works with every compliant gateway — vendor-agnostic by design. Enterprise adds SSO, audit logs, admin controls in Claude.ai.
Gap vs AWS	No managed harness runtime. You must host agents on a partner platform or build the infrastructure yourself.

Open source

LangGraph + Kong Enterprise / WSO2 API Manager

Harness	LangGraph provides graph orchestration, checkpointing, HITL. LangSmith adds observability and eval. You assemble and host everything. Self-assembled
MCP Gateway	Kong Enterprise or WSO2 sit in front of LangGraph as the API gateway — adding auth, rate limiting, routing, and audit logging. Both added MCP adapter support in 2025. Assembled
Memory	You choose: Pinecone, Weaviate, pgvector, Redis — all wired in by your team. No managed default.
Unique strength	Zero vendor lock-in. Run fully on-premises. Maximum control over every component. Best fit for regulated on-prem environments.
Gap vs AWS	Everything the managed platforms give you for free, you build yourself — memory ops, eval pipelines, scaling, HA, audit logs, dashboards.

Agentic Harness products (IBM and Salesforce)

IBM

watsonx.ai + watsonx.governance + API Connect

Harness	watsonx.ai is the agent runtime. watsonx.governance overlays bias detection, drift monitoring, and regulatory compliance mapping on top of every agent run. Compliance-deep
MCP Gateway	IBM API Connect acts as the gateway — OAuth, rate limiting, policy enforcement. MCP adapter added via partner ecosystem. Less native than AWS. Partial
Memory	IBM Db2 and Elasticsearch as structured/search stores. Watson Discovery for unstructured document retrieval.
Unique strength	watsonx.governance is the most mature AI governance overlay in the market — automatically maps every AI decision to EU AI Act, HIPAA, SOC2 requirements.
Gap vs AWS	Harness and gateway are separate products. Smaller MCP ecosystem. Stronger on compliance auditing than on developer experience.

Salesforce

Agentforce + Einstein AI Gateway

Harness	Agentforce is a purpose-built agentic harness embedded inside the Salesforce platform — agents run as first-class objects with CRM data natively accessible. CRM-native
MCP Gateway	Einstein AI Gateway routes model calls and external API calls. MCP support added in 2025. Tightly coupled to Salesforce data model. Partial
Memory	Salesforce Data Cloud is the long-term memory layer — customer records, interaction history, unified profile. The richest CRM-based memory available.
Unique strength	Agents can natively read/write Salesforce records, trigger flows, send emails, and update cases — zero integration required for Salesforce users.
Gap vs AWS	Extremely Salesforce-specific. Limited utility outside the Salesforce ecosystem. Not suitable as a general-purpose agentic platform.

LangGraph and Agentic Harness comparison

Code you write

LangGraph (open source)

You build and operate the full harness yourself

What it is

A Python/JS framework for authoring stateful agent graphs. Nodes, edges, checkpoints, human-in-the-loop — all defined in code. You host it wherever you want.

Effort to start

Medium-high. Learn the graph API, wire your own tools, write state schemas, set up a checkpoint backend (Redis, Postgres), build your own API layer.

Time to production

Weeks to months depending on how much infrastructure you need to build around it.

Cost model

Pay for your own compute (EC2, container, on-prem). No per-call platform fee. Can be cheaper at scale if you tune it.

Control

Total. You define every node, every edge, every retry policy, every data structure. Nothing is hidden or opinionated.

Best for

Regulated environments needing on-prem, custom compliance requirements, or teams that want zero vendor lock-in and have strong engineering resources.

Managed service

Agent harness (AWS AgentCore / Azure AI Foundry / Vertex...)

Vendor runs the infrastructure; you configure and use it

What it is

A fully managed platform that provides orchestration, memory, evaluation, observability, access control, and scaling — all as cloud services you call via API or console.

Effort to start

Low-medium. Connect your agent, configure tool permissions, set evaluation rules. The platform handles the rest operationally.

Time to production

Days to weeks. Most teams get a working production agent in a few days of configuration.

Cost model

Pay per API call, per memory operation, per eval run. Higher unit cost than raw compute, but eliminates engineering overhead cost.

Control

Partial. You configure within the platform's model. Custom logic requires extensions or workarounds. The vendor's opinionated defaults apply.

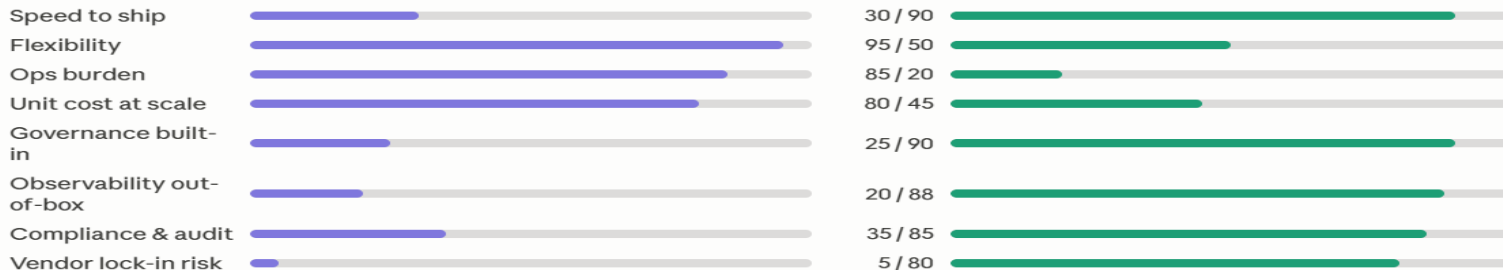
Best for

Teams that want to ship fast, operate in the vendor's cloud already, and don't want to maintain orchestration infrastructure themselves.

LangGraph and Agentic Harness comparison

DIMENSION-BY-DIMENSION SCORE (HIGHER = BETTER ON THAT DIMENSION)

■ LangGraph (hand-coded) ■ Managed harness



WHAT YOU MISS WHEN HAND-CODING — THE GAPS LANGGRAPH ALONE DOES NOT FILL

Memory management

You build short- and long-term stores from scratch. No managed vector DB, no session persistence layer, no TTL eviction — all yours to write and operate.

Evaluation framework

No built-in output scoring or quality gates. You write custom evals or integrate a separate library (LangSmith, RAGAS). They don't run automatically.

Observability & dashboards

No traces, latency metrics, or cost dashboards out of the box. You integrate OpenTelemetry, Datadog, or LangSmith manually — significant setup.

Rate limiting & cost caps

Nothing stops a runaway agent loop from spending \$10k overnight. You must build token budgets and call-rate enforcement yourself.

Secrets & access control

No managed identity layer. You wire IAM roles, secret stores, and per-tool permissions into your code manually — and keep them current.

Scaling & HA

LangGraph runs where you host it. Load balancing, auto-scaling, failover, and zero-downtime deploys are all your infrastructure problem.

Audit log for compliance

No automatic immutable audit trail. HIPAA, SOC2, and CMS audits require you to have built this yourself before an auditor asks for it.

Human-in-the-loop UI

LangGraph supports interrupt points in code, but provides no UI for a human to review and approve. You build that interface separately.

Versioning & rollback

No built-in agent version management. Deploying a new graph version alongside an old one (blue/green, canary) is your deployment pipeline to design.

LangChain and LangGraph capabilities & comparison



Agentic Harness functional representation

